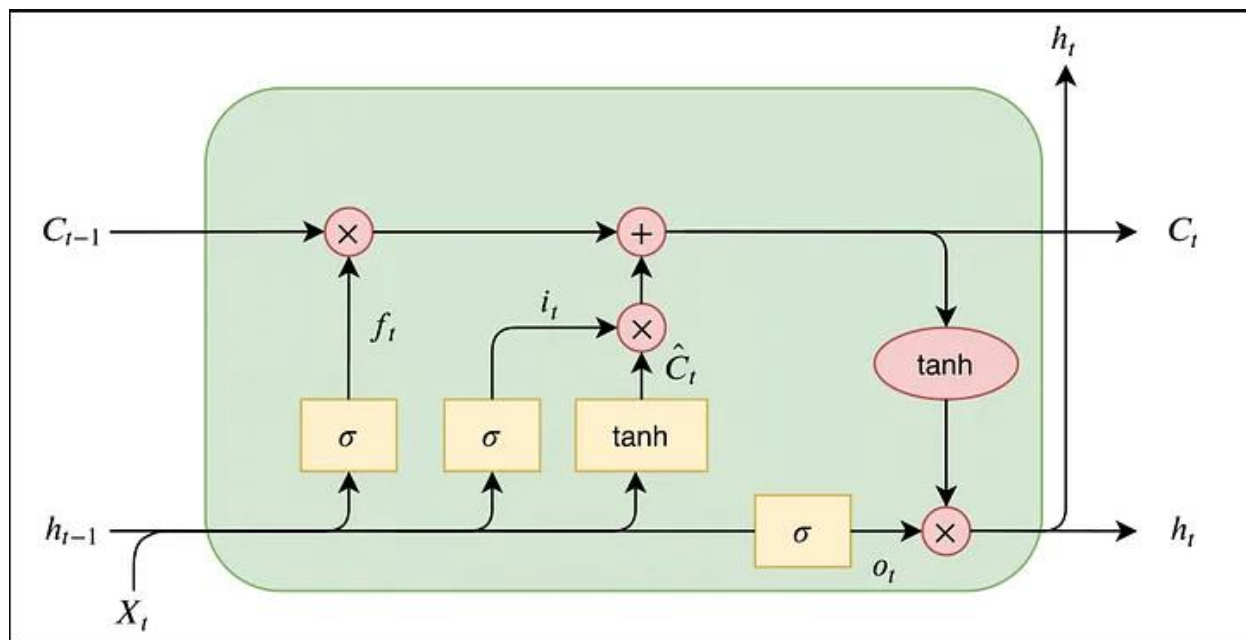


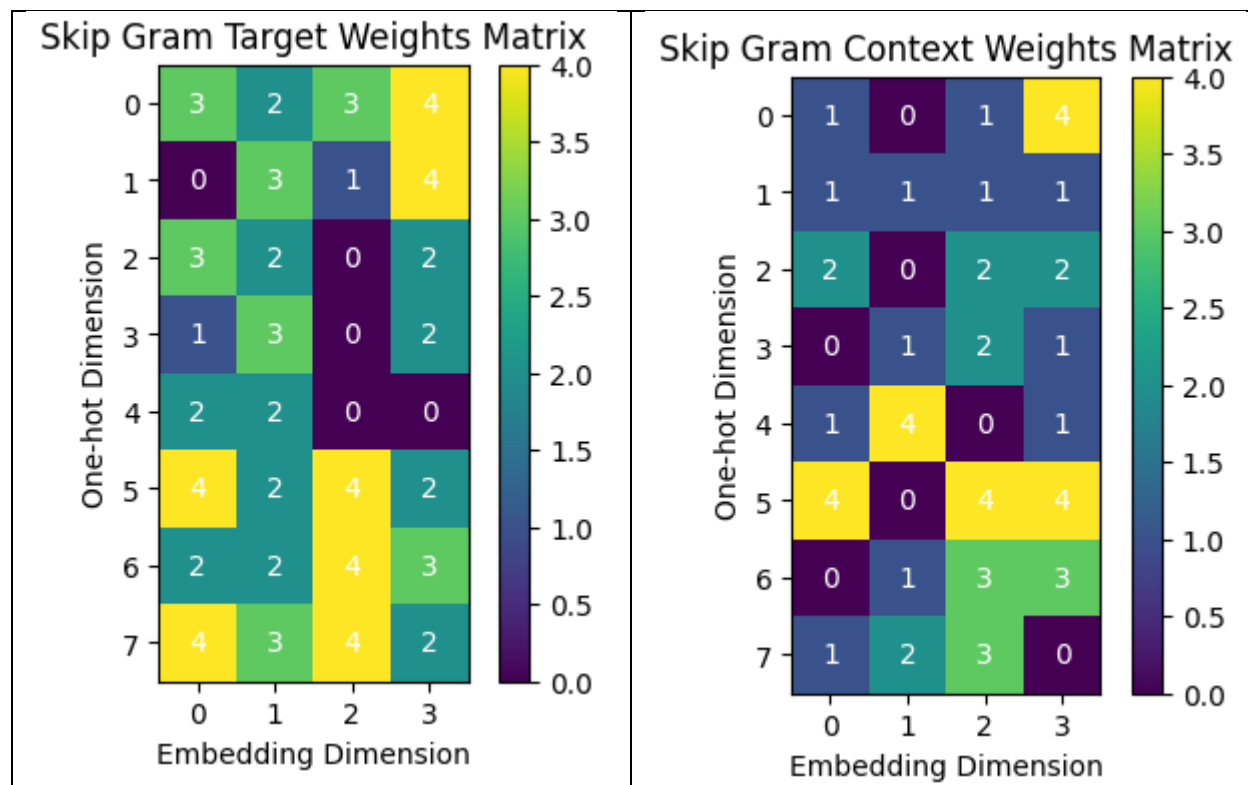
Written Assignment (LSTM)



Vocabulary = {the, students, opened, their, laptops, books, exams, minds}

Task 1: Compute embedding for the given vocabulary using Skip Gram Model.

1. Target Embedding
2. Context Embedding
3. Average Embedding



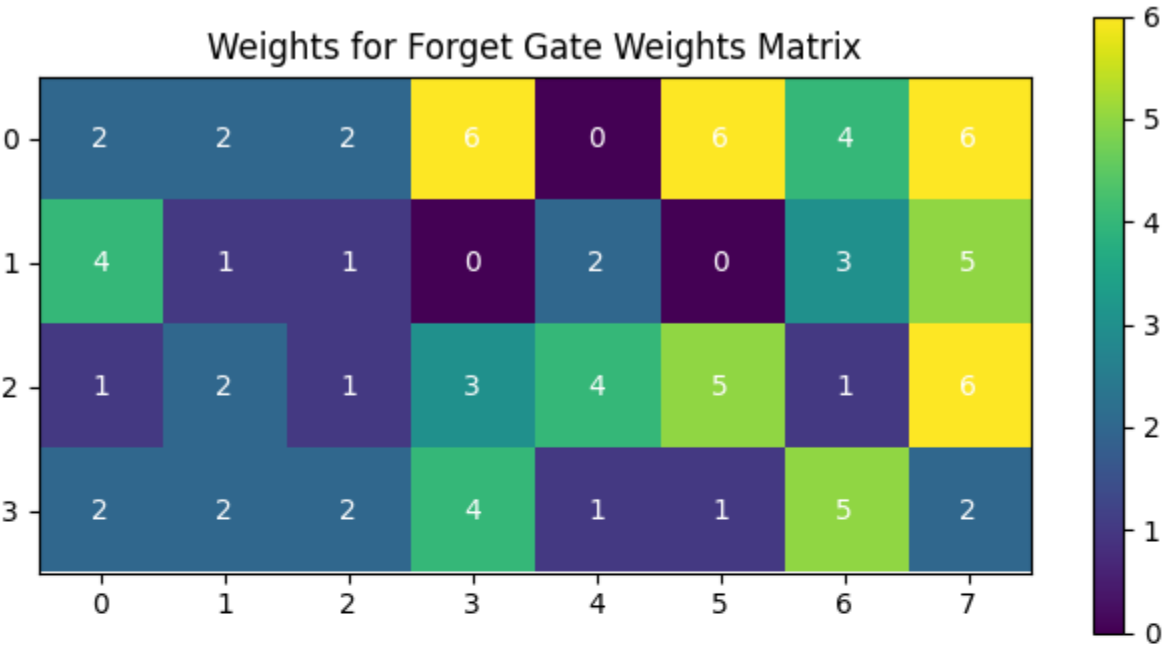
Task 2: Draw the modified architecture for this LSTM Unit and verify if it is the same as mentioned above or if it has an extension.

Task 3: Now utilize embedding for the following sentence to predict the next word in the sequence.

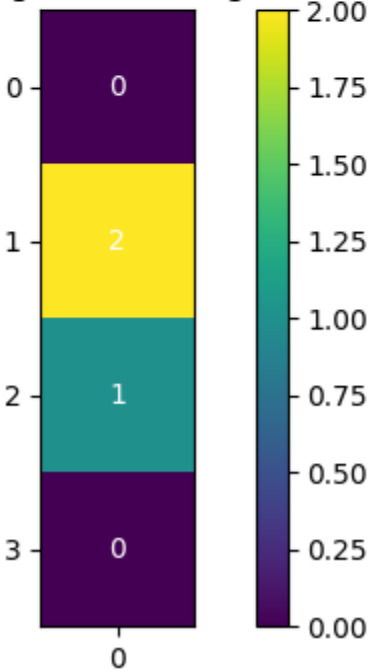
the students opened their _____?

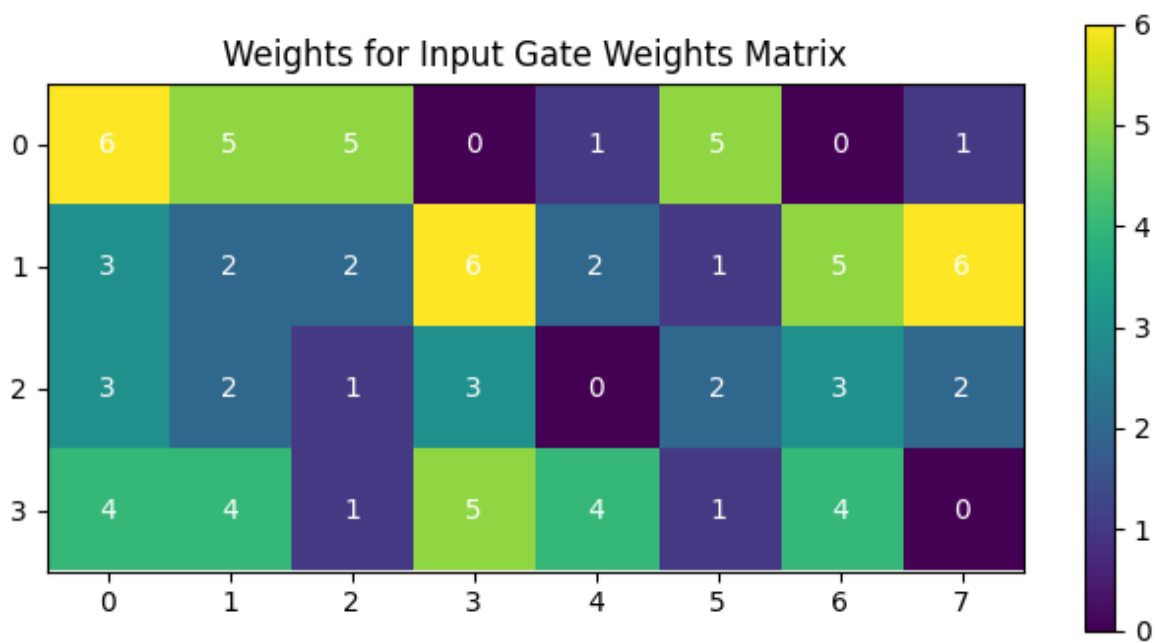
1. You are required to use the given weight matrices to compute the previous cell state and hidden state for each time step.
2. Compute softmax on the output layer to predict probabilities for the next word.
3. Write the word with the highest probability in the form of a one-hot encoded vector and also interpret it from the vocabulary given above.
4. Use all types for embedding in this task and check if using average embedding produces the same results as target embedding and context embedding or not.

Weight and Bias Matrices:

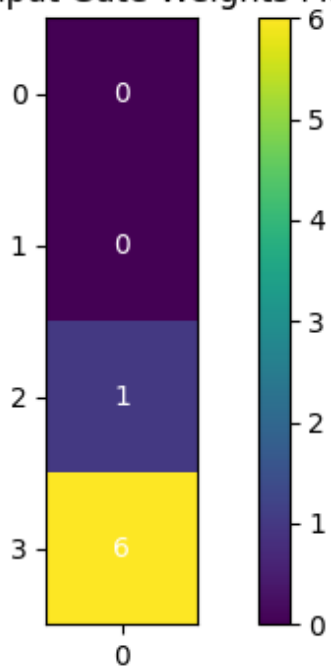


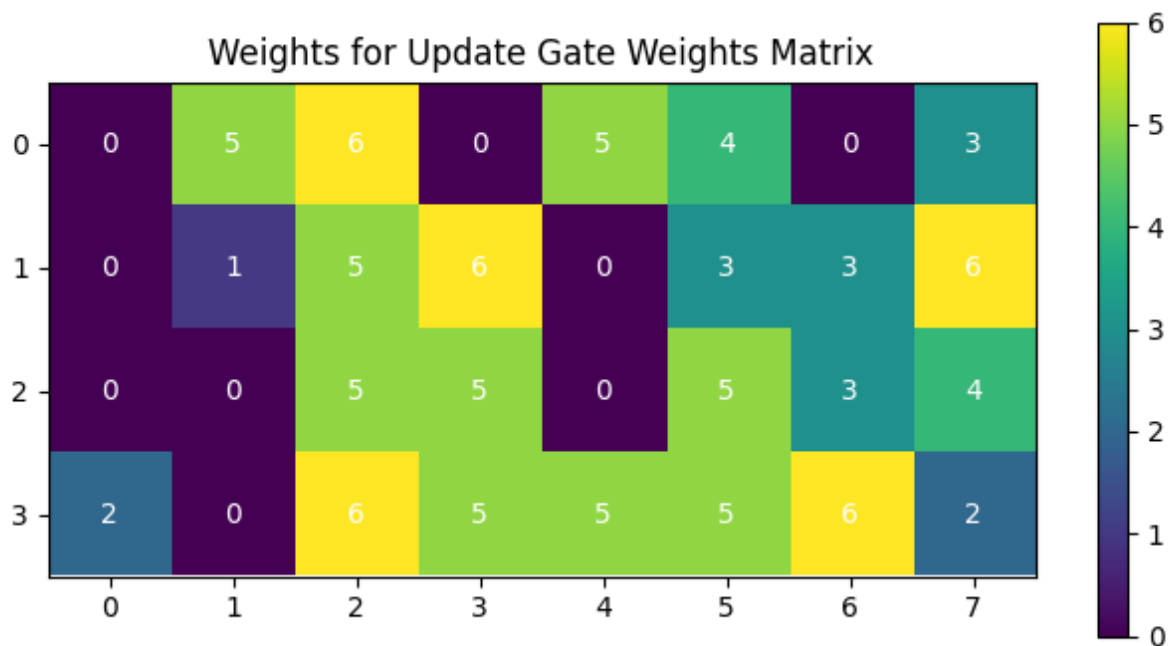
Bias for Forget Gate Weights Matrix



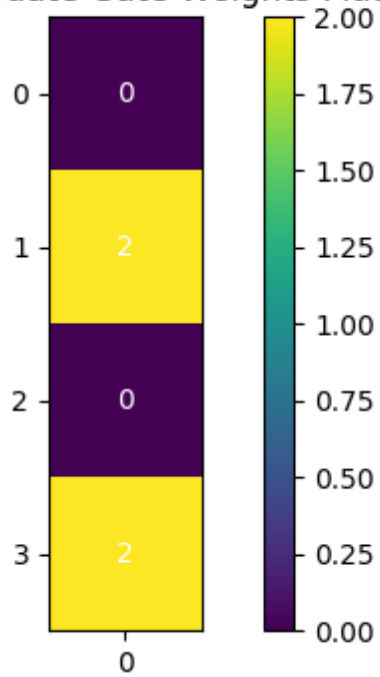


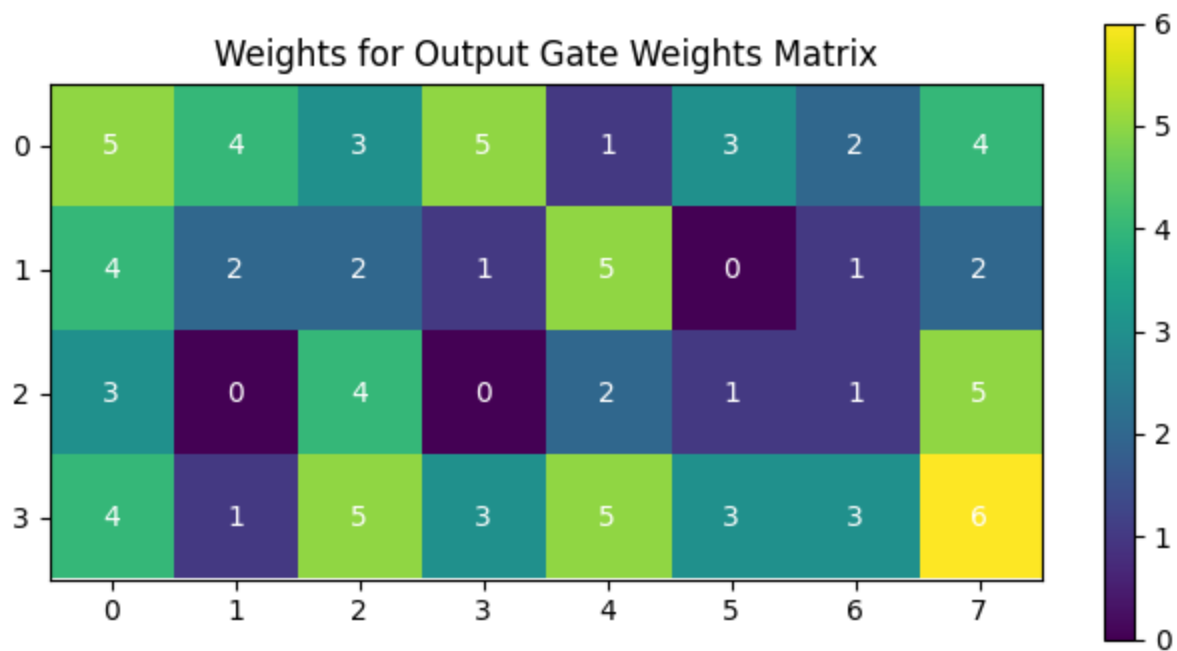
Bias for Input Gate Weights Matrix



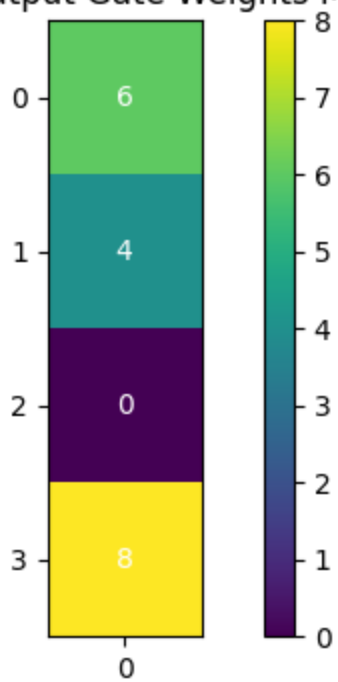


Bias for Update Gate Weights Matrix

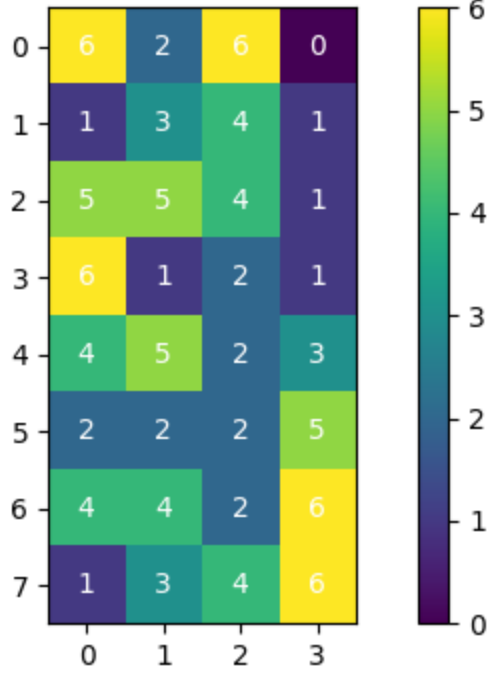




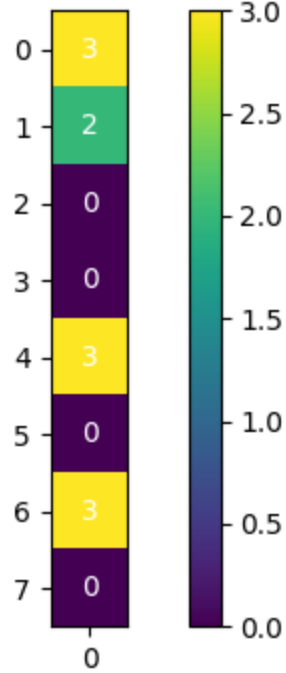
Bias for Output Gate Weights Matrix



Weights for One-Hot Encoded Gate Weights Matrix



Bias for One-Hot Encoded Gate Weights Matrix



Solution:

```
embeddings_target = np.array([[3, 2, 3, 4],
                               [0, 3, 1, 4],
                               [3, 2, 0, 2],
                               [1, 3, 0, 2]])
```

```
Time = 0
Previous Hidden State (ht):
[[0.]
 [0.]
 [0.]
 [0.]]
Previous Cell State (ct):
[[0.]
 [0.]
 [0.]
 [0.]]
Time = 1
Previous Hidden State (ht):
[[0.76159414]
 [0.76159416]
 [0.76159416]
 [0.76159416]]
Previous Cell State (ct):
[[0.99999996]
 [1.         ]
 [1.         ]
 [1.         ]]
Time = 2
Previous Hidden State (ht):
[[0.96402758]
 [0.96402758]
 [0.96402758]
 [0.96402758]]
Previous Cell State (ct):
[[1.99999996]
 [2.         ]
 [2.         ]
 [2.         ]]
Time = 3
Previous Hidden State (ht):
[[0.99505475]
 [0.99505475]
 [0.99505475]
 [0.99505475]]
Previous Cell State (ct):
[[2.99999996]
 [3.         ]
 [2.99999998]
 [2.99999998]]
```


Final Results

Predicted probabilities:

```
[[0.11807027]
 [0.10638317]
 [0.10786921]
 [0.10636569]
 [0.11807027]
 [0.10638313]
 [0.229948  ]
 [0.10691027]]
```

Predicted one-hot encoded vector:

```
[[0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [1.]
 [0.]]
```

```
embeddings_context = np.array([[1, 0, 1, 4],
                                [1, 1, 1, 1],
                                [2, 0, 2, 2],
                                [0, 1, 2, 1]])
```

```
Time = 0
Privous Hidden State (ht):
[[0.]
 [0.]
 [0.]
 [0.]]
```

```
Privous Cell State (ct):
[[0.]
 [0.]
 [0.]
 [0.]]
```

```
Time = 1
Privous Hidden State (ht):
[[0.75876897]
 [0.76159414]
 [0.76159158]
 [0.76159381]]
```

```
Privous Cell State (ct):
[[0.99330715]
 [1.         ]
 [0.99999386]
 [0.99999917]]
```

```
Time = 2
Privous Hidden State (ht):
[[0.96355166]
 [0.96402757]
 [0.96402654]
 [0.96402752]]
```

```
Privous Cell State (ct):
[[1.99330714]
 [1.99999994]
 [1.9999935 ]
 [1.99999911]]
```

```
Time = 3
Privous Hidden State (ht):
[[0.99498828]
 [0.99505475]
 [0.99505469]
 [0.99505474]]
```

```
Privous Cell State (ct):
[[2.99330714]
 [2.99999994]
 [2.9999935 ]
 [2.99999911]]
```

Final Results

Predicted probabilities:

```
[[0.11807004]
 [0.10638315]
 [0.10786918]
 [0.10636567]
 [0.11807027]
 [0.10638311]
 [0.22994831]
 [0.10691026]]
```

Predicted one-hot encoded vector:

```
[[0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [1.]
 [0.]]
```

```
embeddings_average = np.array([[2, 1, 2, 4],
                                [0.5, 2, 1, 2.5],
                                [2.5, 1, 1, 2],
                                [0.5, 2, 1, 1.5]])
```

```
Time = 0
Privous Hidden State (ht):
[[0.]
 [0.]
 [0.]
 [0.]]
Privous Cell State (ct):
[[0.]
 [0.]
 [0.]
 [0.]]
Time = 1
Privous Hidden State (ht):
[[0.76158714]
 [0.76159416]
 [0.76159414]
 [0.76159416]]
Privous Cell State (ct):
[[0.9999833 ]
 [1.         ]
 [0.99999996]
 [1.         ]]
Time = 2
Privous Hidden State (ht):
[[0.9640264 ]
 [0.96402758]
 [0.96402758]
 [0.96402758]]
Privous Cell State (ct):
[[1.9999833 ]
 [2.         ]
 [1.99999996]
 [2.         ]]
Time = 3
Privous Hidden State (ht):
[[0.99505459]
 [0.99505475]
 [0.99505475]
 [0.99505475]]
Privous Cell State (ct):
[[2.9999833 ]
 [3.         ]
 [2.99999995]
 [3.         ]]
```

Final Results

Predicted probabilities:

```
[[0.11807027]
 [0.10638317]
 [0.10786921]
 [0.10636569]
 [0.11807027]
 [0.10638313]
 [0.229948  ]
 [0.10691027]]
```

Predicted one-hot encoded vector:

```
[[0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [1.]
 [0.]]
```